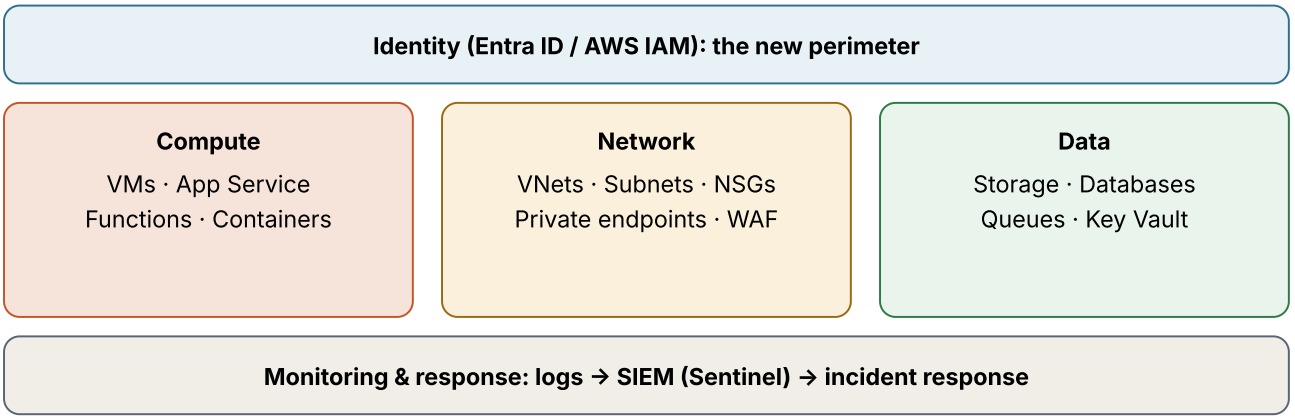
CLOUD SECURITY PATH · STUDY GUIDE 3

Cloud & Cloud Security Foundations

Everything a Cloud Security Engineer builds on: cloud models, compute, storage, networks, databases and queues, identity, data protection, secure design, monitoring and incident response. AWS and Azure side by side, with every service mapped to its twin. Part C covers securing small organisations end to end.



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★ Using this guide with the sprint plan

> **Your plan builds on AWS first (Phase A, Nov 2026 – Feb 2027), adds Azure in sprint S8, then moves to cloud security (Phase B, Mar – Dec 2027). Every table in this guide shows both clouds, so read the AWS column first and the Azure column in S8 and B8.**

Chapters	Read during	Hands-on in (Guide 4)
1–5 Cloud foundations, shared responsibility, accounts	S1	Secure AWS account
8 Networking, 10 Identity	S2	IAM policy + VPC by hand
6 Compute, 7 Storage	S3	EC2 with Session Manager; S3 + CloudFront + WAF
12 Securing cloud infrastructure (IaC)	S4	Terraform
9 Databases and queues	S6	Lambda, DynamoDB, RDS, Cognito
13 Monitoring and logging	S7, B5	CloudTrail → Wazuh; detections
Azure columns throughout	S8	Azure sprint
10–11 Identity and data security (again, deeper)	B1–B2	SCPs, KMS, Secrets Manager, Macie
12 (again), network and posture	B3–B4	WAF, Flow Logs, Config, Prowler, Checkov
14 Cloud incident response	B6	Incident simulations + auto-response
13 KQL section	B8	Sentinel
15–19 Securing small organisations	Alongside, for the startup	Free assessments (Mar – May 2027)

REMEMBER

Wherever a chapter says "your Azure lab", the same exercise works in AWS with the service in the AWS column. Guide 4 has the step-by-step version for each sprint.

1 What cloud computing is

> **Cloud computing is renting computing resources (servers, storage, networks, software) over the internet, on demand, paying only for what you use.**

The standard definition (NIST SP 800-145) lists five essential characteristics:

Characteristic	Meaning	Example
On-demand self-service	Create resources yourself, instantly, without calling anyone	Click "Create VM" in the Azure portal
Broad network access	Reach services over the network from any device	Manage Azure from a laptop or phone
Resource pooling	Many customers share the provider's hardware, isolated from each other	Your VM shares a physical host with strangers' VMs
Rapid elasticity	Scale up or down quickly, even automatically	Add 10 web servers during a sale, remove them after
Measured service	Usage is metered and billed	Pay per second of VM time and per GB stored

THINK OF IT LIKE THIS

Owning servers is like owning a generator: you buy it, fuel it and fix it. Cloud is like the electricity grid: plug in, use what you need, pay the bill. The catch is that you're still responsible for what you plug in and who you let use the sockets.

IN YOUR LAB

Your home lab is a small private "cloud" on one laptop: resource pooling (four VMs sharing hardware), but no self-service portal, elasticity or billing.

QUICK CHECK (answers at the back)

1. Name the five essential characteristics of cloud computing.
2. Which characteristic means many customers share the same hardware?

2 Service models: IaaS, PaaS, SaaS

> **The service model decides how much the provider manages and how much you manage. The more the provider manages, the less you control, and the less you must secure yourself.**

Model	You get	You manage	Azure	AWS
IaaS: Infrastructure as a Service	Raw VMs, disks, networks	OS, patching, apps, data, access	Virtual Machines	EC2
PaaS: Platform as a Service	A ready platform to run your code or database	Your code, data, access, configuration	App Service, Azure SQL	Elastic Beanstalk, RDS
SaaS: Software as a Service	A finished application	Your data, users and settings	Microsoft 365	(third-party SaaS on AWS)
FaaS / serverless	Run individual functions on events	Function code and permissions	Azure Functions	Lambda
BaaS: Backend as a Service	Ready-made app backends: login, database, file storage	App logic, data rules	(Azure Mobile/App Service features)	Amplify

THINK OF IT LIKE THIS

Pizza as a service. IaaS: you're given a kitchen and you cook. PaaS: the pizza is baked for you; you choose the toppings. SaaS: the pizza is delivered and you just eat. In every case, *you* decide who gets a slice, which is identity and data.

REMEMBER

Whatever the model, **you always own your data, your identities and your access settings**. Most real cloud breaches happen in that customer-owned layer.

QUICK CHECK (answers at the back)

3. In IaaS, who patches the operating system?
4. Give one Azure and one AWS example of PaaS.
5. What do you still own even in SaaS?

3 Deployment models

> **The deployment model says who owns the infrastructure and who can use it.**

Model	What it is	Pros	Cons	Example
Public	Provider-owned, shared by many customers	No hardware to buy; global scale; pay as you go	Less control; shared tenancy; costs can surprise you	Azure, AWS, Google Cloud
Private	Dedicated to one organisation, on-prem or hosted	Full control; data stays in-house	Expensive; you run it all	A bank's VMware or OpenStack data centre; your home lab
Hybrid	Private and public connected and used together	Keep sensitive systems in-house, burst to the cloud	Complex networking and identity	On-prem AD synced to Entra ID; **your lab v2**
Community	Shared by organisations with common needs	Shared cost; common compliance	Shared governance	Government clouds, healthcare consortia
Multi-cloud	Using more than one public cloud	Avoids lock-in; best service from each	Several security models to learn	Azure for identity, AWS for apps

IN YOUR LAB

Lab v2 is **hybrid**: your on-prem lab (private) connected to Azure (public) through Azure Arc, monitored by one SIEM. That's what most companies really run.

QUICK CHECK (answers at the back)

6. What makes a cloud "hybrid"?
7. Give one reason a bank might keep a private cloud.

4 The shared responsibility model

> **Security *of* the cloud is the provider's job. Security *in* the cloud is your job. Where the line falls depends on the service model.**

IaaS	PaaS	SaaS
Data & access	Data & access	Data & access
Applications	Applications	Applications
Operating system	Operating system	Operating system
Network controls (NSGs)	Network controls (shared)	Network controls
Virtualization	Virtualization	Virtualization
Physical hosts & network	Physical hosts & network	Physical hosts & network
Data centre	Data centre	Data centre

You (customer) · Provider · Shared

Always yours, in every model: your data, your identities and accounts, who has access, and how you configure what you're given.

REMEMBER

Most cloud breaches aren't the provider being hacked. They're customer-side mistakes: **public storage, leaked keys, over-permissive access, missing MFA**. A Cloud Security Engineer's job is mostly preventing and catching those.

QUICK CHECK (answers at the back)

8. In PaaS, who patches the operating system?
9. Name two security responsibilities that stay with the customer in every model.

5 Global infrastructure and accounts

> **Clouds are made of regions (geographic areas), each with several availability zones (separate data centres). Your resources live inside an account structure that also defines security boundaries.**

Concept	Azure	AWS	Why it matters
Geographic area	Region (e.g. South Africa North, UK South)	Region (e.g. af-south-1, eu-west-2)	Latency, cost and **data residency law**
Separate data centres	Availability zones	Availability zones	Survive a data-centre failure
Top of the hierarchy	Entra tenant → management groups	AWS Organizations → organizational units	Apply policies to many accounts at once
Billing and security boundary	Subscription	Account	Separate dev, prod and lab environments
Grouping of resources	Resource group	(tags, CloudFormation stacks)	Manage and delete things together
The all-powerful identity	Global Administrator; subscription Owner	**Root user**	Protect with MFA; never use day to day

First things to do in any new cloud account

1. **MFA** on the top account (Azure Global Admin; AWS root user).
2. Create a separate admin user for daily work. Lock the top account away and keep a "break-glass" emergency account.
3. Set a **budget with alerts** before you build anything.
4. Turn on the activity logs (on by default in Azure; in AWS, create a CloudTrail trail that keeps logs long-term).
5. Decide naming and **tags** (owner, environment, cost centre).

QUICK CHECK (answers at the back)

10. What is the AWS equivalent of an Azure subscription?
11. What two things should you set up before building anything in a new cloud account?

6 Compute

> **Compute is where code runs: full VMs you manage yourself, managed app platforms, serverless functions or containers.**

Type	Azure	AWS	You secure
Virtual machines (IaaS)	Azure Virtual Machines, scale sets	EC2, Auto Scaling groups	OS hardening, patching, NSG/SG rules, disk encryption, access (no public SSH/RDP)
Managed web apps (PaaS)	**App Service**	**Elastic Beanstalk**	App code, HTTPS-only, authentication, secrets, network restrictions
Serverless functions	Azure Functions	Lambda	Function permissions (least privilege), input validation, secrets
Containers	Container Instances, Container Apps, AKS	ECS, Fargate, EKS	Image security, registry access, runtime privileges (Guide 2)

VM security checklist

- **No public IP**, or none with SSH/RDP open to the internet. Use Azure Bastion or AWS Session Manager.
- **SSH keys**, not passwords (as on your Ubuntu server). Microsoft Entra login for Windows and Linux VMs where possible.
- **Patch automatically** (Azure Update Manager; AWS Systems Manager Patch Manager).
- **Managed identity / IAM role** for the VM, never access keys stored on the disk.
- **Encrypt disks** (on by default; customer-managed keys for stricter control).
- **Endpoint protection** (Defender for Servers; or the Wazuh agent you already know).

App Service vs Elastic Beanstalk

Both let you deploy a web app without managing servers. **App Service** is fully managed PaaS: you upload code and Azure runs it. **Elastic Beanstalk** is an orchestration layer: it creates and manages EC2, load balancers and scaling *for* you, and you can still see and change those resources.

QUICK CHECK (answers at the back)

12. Why should a VM use a managed identity or IAM role instead of an access key?
13. How do you reach a VM securely with no public IP?

7 Storage

> **Cloud object storage holds files ("objects" or "blobs") at huge scale. It's one of the most useful cloud services, and historically one of the most common sources of data leaks.**

	Azure	AWS
Object storage	**Storage account → containers → blobs**	**S3 buckets → objects**
File shares	Azure Files	EFS, FSx
VM disks	Managed disks	EBS volumes
Cost tiers	Hot, Cool, Cold, Archive	Standard, Infrequent Access, Glacier classes
Block public access	Disable "allow blob public access"	**S3 Block Public Access**
Temporary access link	**SAS** (shared access signature)	**Pre-signed URL**
Ransomware protection	Soft delete, versioning, immutable storage	Versioning, **Object Lock**
Private network access	Private endpoint	VPC endpoint (gateway / interface)

Storage security essentials

- **Never public by default.** Keep public access blocked at the account level, and allow exceptions only deliberately.
- **Prefer identity-based access** (Entra ID / IAM roles) over shared keys. Azure storage **account keys** give full access, so treat them like root passwords and rotate them.
- **Short-lived SAS / pre-signed URLs**: minimal permissions and a short expiry. A long-lived SAS link is a leaked password waiting to happen.
- **Encryption at rest** is on by default in both clouds. Use customer-managed keys when policy requires it (chapter 11).
- **Enforce HTTPS / TLS** for all access.
- **Turn on logging** (Azure storage diagnostic logs; S3 server access logs or CloudTrail data events) to see who read what.
- **Versioning plus immutability**, so ransomware or a malicious admin can't silently destroy data.

QUICK CHECK (answers at the back)

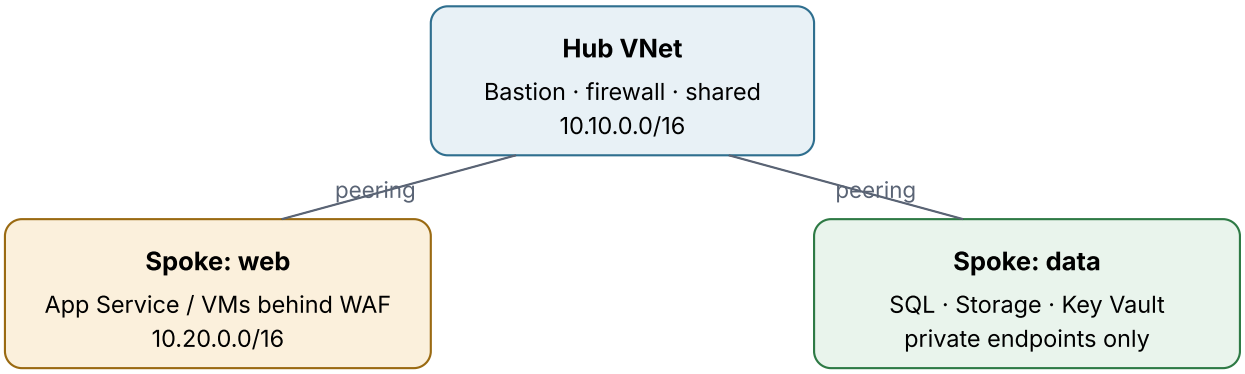
14. What is the Azure equivalent of an S3 bucket?
15. Why are long-lived SAS links risky?
16. Name two controls that protect stored data from ransomware.

8 Networking: VNets and VPCs

> **A VNet (Azure) or VPC (AWS) is your private network in the cloud, split into subnets and controlled by route tables and network security rules. It's your pfSense lab, rebuilt in software.**

Concept	Azure	AWS	Your lab
Private network	VNet	VPC	The whole lab
Segment	Subnet	Subnet (in one AZ)	LAN / DMZ / MGMT
Routing	Route tables (system routes plus user-defined routes)	Route tables	pfSense routing
Internet in	Public IP / load balancer	Internet gateway + public IP	None (by design)
Internet out only	NAT gateway	NAT gateway	pfSense outbound NAT
Stateful filtering	**NSG** (subnet or NIC), plus application security groups	**Security group** (instance/ENI, allow rules only)	pfSense rules
Stateless filtering	(none)	**Network ACL** (subnet level, allow and deny)	
Connect networks	VNet peering, VPN gateway, ExpressRoute	VPC peering, Transit Gateway, VPN, Direct Connect	
Private access to PaaS	Private endpoint (Private Link)	VPC interface endpoint (PrivateLink)	
Admin access	Azure Bastion	Session Manager	SSH from Kali
Central firewall / WAF	Azure Firewall; App Gateway WAF	AWS Network Firewall; AWS WAF	pfSense
Traffic logs	VNet / NSG flow logs	VPC Flow Logs	pfSense firewall logs

A secure design: hub and spoke



- Shared security services (Bastion, firewall) live in the **hub**. Workloads live in **spokes**, peered to the hub.
- The data spoke has **no internet path**. Databases and storage are reached only through **private endpoints**.
- NSGs on every subnet, with **default deny** and only the flows each tier needs, exactly like your pfSense zones.

REMEMBER

Two cloud-specific traps. **(1)** Subnets in the same VNet/VPC can reach each other **by default**, so segmentation must be added with NSGs or security groups. **(2)** A "private" PaaS service may still have a **public endpoint** unless you disable it.

QUICK CHECK (answers at the back)

17. Security group vs network ACL: which is stateful?
18. What is a private endpoint for?
19. Why put Bastion and the firewall in a hub VNet?

9 Databases and queues

> **Managed databases and messaging services are PaaS: the provider runs the servers, and you secure access, network exposure, encryption and auditing.**

Databases

Type	Azure	AWS
Relational (SQL)	Azure SQL Database, Azure Database for PostgreSQL / MySQL	RDS (PostgreSQL, MySQL, SQL Server…), Aurora
NoSQL	Cosmos DB	DynamoDB
Cache	Azure Cache for Redis	ElastiCache

Secure a database by: disabling public network access (use private endpoints); using Entra ID / IAM authentication instead of shared passwords; encryption at rest (TDE, on by default) and TLS in transit; automated backups with tested restores; auditing and threat detection (Defender for SQL; RDS logs to CloudWatch); least-privilege database users for applications.

Queues and messaging

Queues let parts of an application hand work to each other without waiting: an order is placed in a queue, and a worker processes it later. This makes apps resilient and scalable.

Purpose	Azure	AWS
Simple queue	**Queue Storage**	**SQS**
Enterprise messaging (queues and topics)	Service Bus	SQS + **SNS**, Amazon MQ
Events	Event Grid, Event Hubs	EventBridge, Kinesis

Secure queues by: using managed identities / IAM roles for producers and consumers; private endpoints; encryption; dead-letter queues for poisoned messages; and treating message content as **untrusted input** that gets validated before use.

QUICK CHECK (answers at the back)

20. What is the AWS equivalent of Azure Queue Storage?
21. Give three controls for securing a managed database.

10 Identity and access management

> **In the cloud, identity is the perimeter. Anyone with valid credentials and permissions can reach your resources from anywhere, so IAM is the most important cloud security skill.**

The building blocks

Concept	Azure / Entra ID	AWS
Directory of identities	**Microsoft Entra ID** (users, groups)	IAM users and groups; IAM Identity Center for SSO
Identity for apps	Service principal; **managed identity**	**IAM role** (temporary credentials via STS)
Permissions	**Azure RBAC**: role + identity + scope	**IAM policies** (JSON) attached to identities or resources
Top-level guardrails	Azure Policy; management groups	Service control policies (SCPs)
Risk-based sign-in rules	**Conditional Access** (Entra ID P1); security defaults (free)	IAM conditions; IAM Identity Center MFA
Just-in-time admin	**Privileged Identity Management (PIM)** (P2)	Temporary role assumption, time-bound access

Azure RBAC: who, what, where

A **role assignment** = a **security principal** (who) + a **role definition** (what: e.g. Owner, Contributor, Reader) + a **scope** (where: management group → subscription → resource group → resource). Permissions inherit downwards. Note that **Entra roles** (like Global Administrator, which manage the directory) are separate from **Azure RBAC roles** (which manage resources).

An AWS IAM policy

```
{
  "Version": "2012-10-17",
  "Statement": [{
    "Effect": "Allow",
    "Action": ["s3:GetObject"],
    "Resource": "arn:aws:s3:::lab-reports/*"
  }]
}
```

This allows **only** reading objects in one bucket. Anything not allowed is **implicitly denied**, and an explicit `"Deny"` always wins. That's the same default-deny logic as your pfSense rules.

IAM best practices

- **MFA everywhere**, phishing-resistant (FIDO2 keys, passkeys) for admins.
- **Least privilege:** start with nothing, add only what's needed; prefer narrow scopes and Reader over Contributor.

- **No long-lived keys:** use managed identities and roles. Never put access keys in code or Git.
- **Separate admin accounts** and just-in-time elevation (PIM).
- **Block legacy authentication** (protocols that can't do MFA).
- **Review access regularly;** remove leavers and unused permissions.
- **Break-glass accounts:** two emergency admin accounts, excluded from normal policies, strongly protected and monitored.

REMEMBER

The most common cloud attack chain: **phished or leaked credential → no MFA → over-privileged account → data stolen or crypto-miners deployed**. Every IAM control breaks one link.

QUICK CHECK (answers at the back)

22. What three parts make an Azure role assignment?
23. In AWS IAM, what happens to an action that no policy allows?
24. Why prefer a managed identity over an access key?

11 Data security

> **Data must be protected in all three states: at rest (stored), in transit (moving) and in use (being processed).**

State	Threat	Controls	Azure	AWS
At rest	Stolen disks, snapshots or backups; storage misconfiguration	Encryption, access control, immutability	Storage Service Encryption, Azure Disk Encryption, SQL TDE, **Key Vault**	S3/EBS/RDS encryption, **KMS**
In transit	Eavesdropping, man-in-the-middle	TLS 1.2+, HTTPS-only, private connections, VPN	HTTPS-only settings, minimum TLS version, Private Link	ACM certificates, bucket policies requiring TLS, PrivateLink
In use	Memory scraping, a malicious admin on the host	Confidential computing (hardware-isolated enclaves)	Confidential VMs and containers	Nitro Enclaves

Key management

- **Platform-managed keys** (the default): the provider creates and rotates keys. Easy, and fine for most data.
- **Customer-managed keys (CMK)** in Key Vault or KMS: you control rotation and can revoke access. Needed for some compliance rules.
- Separate who can **use** keys from who can **manage** them (separation of duties); log every key operation; enable soft delete and purge protection so keys can't be destroyed by mistake or by an attacker.
- **Secrets** (passwords, API keys, connection strings) go in Key Vault or AWS Secrets Manager, never in code or config files.

Knowing what you're protecting

- **Data classification:** label data as public, internal, confidential or restricted, then apply controls by label (Microsoft Purview; Amazon Macie finds sensitive data in S3).
- **Data loss prevention (DLP):** block sensitive data leaving by email, sharing or USB.
- **Backups:** encrypted, immutable and tested (Guide 2, chapter 6).
- **Data residency:** privacy laws may require personal data to stay in certain countries. Choose regions accordingly.

QUICK CHECK (answers at the back)

25. Name the three states of data and one control for each.
26. What is a customer-managed key, and why use one?
27. Where should application secrets be stored?

12 Securing cloud infrastructure

> **Secure cloud infrastructure is designed in, defined as code, enforced by policy and checked continuously, not configured by hand and hoped for.**

The building blocks

Practice	What it means	Azure	AWS
Landing zone	A pre-built, secure foundation: account structure, networks, identity, logging, policies	Azure Landing Zones	Control Tower
Infrastructure as Code (IaC)	Define resources in code files, review them like software, and deploy them repeatably	Terraform, Bicep	Terraform, CloudFormation
Policy as code	Rules that block or flag insecure settings automatically	Azure Policy	AWS Config rules, SCPs
Posture management (CSPM)	Continuously scan for misconfigurations and score your security	Defender for Cloud (secure score)	Security Hub
Workload protection (CWPP)	Threat protection for VMs, containers, databases	Defender for Cloud plans	GuardDuty, Inspector
Benchmarks	Agreed secure baselines to measure against	Microsoft cloud security benchmark, CIS	AWS Foundational Security Best Practices, CIS

The misconfigurations behind most cloud incidents

1. Public storage buckets or containers.
2. SSH/RDP (22/3389) open to the internet (`0.0.0.0/0`).
3. Admin accounts without MFA; the root account used day to day.
4. Access keys in code, Git repos or VM disks.
5. Over-permissive roles (Owner or `*:*` everywhere).
6. Databases with public endpoints.
7. Logging turned off or not retained.
8. Unencrypted or publicly shared snapshots.

Why IaC is a security control

With Terraform, your whole environment is a reviewable file. You can **scan it before deployment** (tools such as Checkov or tfsec flag insecure settings), keep it in Git (who changed what, and when), rebuild it identically after an incident, and **destroy it at the end of every lab session** to avoid costs.

IN YOUR LAB

Phase 8 builds the hub-and-spoke network in Azure with Terraform. Phase 12 scans AWS with **Prowler**, an open-source posture scanner, and fixes the findings.

QUICK CHECK (answers at the back)

28. What does a CSPM tool do?
29. Name four common cloud misconfigurations.
30. Give two ways IaC improves security.

13 Monitoring and logging

> **You can't detect what you don't log. Cloud logs fall into three groups: control-plane activity (who changed what), sign-ins, and data or network activity. A SIEM brings them together.**

What you need to know	Azure	AWS
Who did what to my resources? (control plane)	**Activity Log**	**CloudTrail** (management events)
Who signed in, from where, and did it succeed?	Entra ID **sign-in** and **audit** logs	CloudTrail (console logins), IAM Identity Center logs
Who read or wrote my data?	Resource / diagnostic logs (storage, SQL, Key Vault)	CloudTrail data events, S3 access logs
What network traffic flowed?	VNet / NSG flow logs	**VPC Flow Logs**
Metrics, log storage, alerts	Azure Monitor, **Log Analytics workspace**	**CloudWatch** (metrics, Logs, alarms)
Automatic threat detection	**Defender for Cloud**	**GuardDuty** (analyses CloudTrail, VPC flow and DNS logs)
Configuration history and compliance	Azure Policy, Resource Graph	AWS Config
Central findings dashboard	Defender for Cloud	Security Hub
SIEM + SOAR	**Microsoft Sentinel**	(Security Lake + partner SIEMs, or Sentinel connectors)

KQL: the language of Sentinel and Log Analytics

This query looks for many failed sign-ins per user and IP in 5-minute windows. It's the cloud version of your Wazuh rule 100100:

```
SigninLogs
| where ResultType != "0"            // "0" means success
| summarize Failures = count() by UserPrincipalName, IPAddress, bin(TimeGenerated, 5m)
| where Failures > 10
| order by Failures desc
```

Read it top to bottom: take the sign-in logs → keep failures → count them per user, IP and 5-minute window → keep only the noisy ones → sort.

Logging essentials

- Turn on logs **before** you need them, because you can't go back in time.
- Send logs to a **central workspace** in a separate, locked-down subscription or account, so an attacker who compromises a workload can't delete the evidence.
- Set **retention** to match policy and law.

- Keep clocks consistent. The cloud does this for you, but hybrid sources need NTP (your Case 001).
- Tune alerts and remove duplicates, as you did in your lab.

QUICK CHECK (answers at the back)

31. Which service answers "who changed my resources?" in Azure, and which in AWS?
32. What does GuardDuty analyse?
33. Why store logs in a separate, locked-down subscription or account?

14 Cloud incident response

> **Incident response in the cloud follows the same lifecycle as on-prem, but evidence lives in logs and APIs, attackers move through identities, and containment is often a configuration change.**

The lifecycle (NIST SP 800-61)

1. **Preparation:** logging on, a SIEM, playbooks, break-glass accounts, contacts, and practice.
2. **Detection and analysis:** triage the alert, scope it, build a timeline from logs.
3. **Containment, eradication and recovery:** stop the spread, remove the attacker, restore safely.
4. **Post-incident activity:** lessons learned, fix the root cause, update detections.

Four common cloud incidents

Scenario	Contain	Investigate	Recover and prevent
Leaked access key (e.g. pushed to GitHub)	Deactivate the key immediately; create a replacement	CloudTrail / Activity Log for everything the key did: new users, new keys, new resources, data access	Remove anything the attacker created; secret scanning on repos; use roles instead of keys
Compromised user account	Disable the account or reset its password; **revoke sessions and tokens**; re-register MFA	Sign-in logs (locations, IPs), audit logs, mailbox rules and forwarding, app consents	Enforce MFA and Conditional Access; block legacy authentication
Public storage exposure	Block public access	Access logs: was data actually downloaded, by whom?	Account-level public-access block; policy to deny public storage; legal notification if personal data was exposed
Compromised VM (e.g. a crypto-miner)	**Isolate** with a quarantine NSG / security group; **don't delete it yet**	**Snapshot the disk** for evidence; review flow logs, process lists, and the identity the VM used	Rebuild from a clean image; rotate any credentials the VM could reach; patch the entry point

Automation (SOAR)

Sentinel playbooks (built on Logic Apps) and **AWS EventBridge + Lambda** can respond in seconds: disable a user, block an IP, isolate a VM, or open a ticket and notify the analyst. Automate the repetitive, well-understood steps, and keep a human in the loop for decisions that are destructive or ambiguous.

REMEMBER

In the cloud, **preserve before you destroy**. Snapshot disks and export logs before deleting anything. A deleted VM takes its evidence with it.

QUICK CHECK (answers at the back)

34. What are the four phases of the NIST incident response lifecycle?
35. A developer leaked an access key on GitHub. What is your first action?
36. Why snapshot a compromised VM before deleting it?

15 Securing small organisations: the essential controls

> **Most small businesses don't need complex tools. A short list of basics, done consistently, stops the majority of real attacks: phishing, account takeover, unpatched systems and ransomware.**

These follow the spirit of the **CIS Critical Security Controls, Implementation Group 1** ("essential cyber hygiene") and similar national baselines.

#	Control	What "good" looks like	Tools
1	**Know your assets**	A list of every laptop, phone, server, account and cloud service	Wazuh inventory (syscollector), Intune, RMM tools
2	**MFA everywhere**	Email, admin accounts, remote access and finance systems all need MFA	Entra ID security defaults or Conditional Access
3	**Patch promptly**	OS and apps updated automatically; critical fixes within days	Windows Update / Intune, Wazuh vulnerability detection
4	**Endpoint protection**	Antivirus/EDR on every device, monitored	Microsoft Defender, other EDR products, the Wazuh agent
5	**Email security**	Spam and phishing filtering; SPF, DKIM and DMARC set up	Microsoft 365 / Google Workspace protections (chapter 16)
6	**Backups**	3-2-1, one copy immutable or offline, restores tested	Azure Backup, M365 backup tools, external drives
7	**Least privilege**	Staff don't use admin accounts day to day; leavers removed promptly	Entra ID, local admin removal
8	**Security awareness**	Short, regular phishing and scam training; a way to report suspicious messages	Simulated phishing tools, short briefings
9	**Logging and monitoring**	Key logs collected and someone actually reviews alerts	Wazuh, Sentinel
10	**Incident response plan**	A one-page plan: who to call, what to do, how to restore	A written plan, tested once a year

REMEMBER

For small organisations, the highest-value controls are usually **MFA, patching, tested backups and email security**. Get those right before buying anything advanced.

QUICK CHECK (answers at the back)

37. Name the four highest-value controls for a small organisation.
38. What does "least privilege" mean for everyday staff accounts?

16 Email and Microsoft 365 security

> **Email is the front door for most attacks on businesses: phishing, malware and business email compromise (BEC), where an attacker takes over or impersonates a mailbox to redirect payments.**

Email authentication: SPF, DKIM, DMARC

Record	What it proves	Example DNS TXT record
SPF	Which servers may send email for your domain	`v=spf1 include:spf.protection.outlook.com -all`
DKIM	The message was signed by your domain and not altered in transit	A public key published in DNS; the mail system signs each message
DMARC	What receivers should do when SPF/DKIM fail, and where to send reports	`v=DMARC1; p=quarantine; rua=mailto:dmarc@example.com`

Roll out DMARC gradually: start with `p=none` to collect reports, fix legitimate senders, then move to `p=quarantine` and finally `p=reject`. Check any domain's records with `dig TXT example.com` and `dig TXT _dmarc.example.com`.

Microsoft 365 / Entra ID baseline

- **MFA for everyone:** security defaults (free) or Conditional Access policies (Entra ID P1).
- **Block legacy authentication**, which bypasses MFA.
- **Separate admin accounts**, with as few Global Administrators as possible (but at least two, including a break-glass account).
- **Defender for Office 365** protections such as Safe Links and Safe Attachments, where licensed.
- **Disable automatic external forwarding**, a classic BEC trick for quietly copying mail out.
- **Turn on audit logging**, and watch for suspicious **inbox rules** (e.g. rules that move payment emails to a hidden folder).
- Control **app consent**, so users can't grant risky third-party apps access to mail and files.

REMEMBER

BEC signs: a sign-in from an unusual country, a new inbox rule, email forwarding to an outside address, and then a "change of bank details" message. Alerting on those catches many attacks early.

QUICK CHECK (answers at the back)

39. What do SPF, DKIM and DMARC each do?
40. Why roll DMARC out starting with p=none?
41. Name two signs of business email compromise.

17 Vulnerability management, patching and backups

> **Vulnerability management is a cycle: find weaknesses, decide which matter most, fix them, and prove they're fixed.**

The cycle

1. **Discover:** keep an inventory and scan systems for known vulnerabilities (CVEs).
2. **Prioritise:** not by score alone. Combine the **CVSS** severity with whether it's **actively exploited** (CISA's Known Exploited Vulnerabilities list is a good signal), whether the system is exposed to the internet, and how important it is to the business.
3. **Remediate:** patch, upgrade, or apply a workaround (disable the feature, restrict network access).
4. **Verify:** rescan to confirm the fix.
5. **Report:** show the trend over time.

Tool	Type	Notes
Wazuh vulnerability detection	Agent-based, free	Uses the software inventory your agents already collect
Wazuh SCA (configuration assessment)	Agent-based, free	Checks settings against CIS benchmarks
Greenbone / OpenVAS	Network scanner, open source	Scans hosts over the network
Defender Vulnerability Management	Commercial	Built into Defender for Endpoint and Defender for Servers
Amazon Inspector	Commercial (AWS)	Scans EC2, container images and Lambda

Only scan systems you own or have written permission to test. For a client, that permission must be part of the agreement.

Patching in practice

- Enable automatic updates for operating systems and browsers.
- Agree a cadence: critical and actively exploited fixes within days, others monthly.
- Test on a few machines first, then roll out.
- Track what can't be patched (old systems) and isolate it.

Backups, revisited

Follow 3-2-1 with at least one immutable or offline copy (Guide 2, chapter 6). Back up Microsoft 365 or Google Workspace data too: SaaS providers keep the service running, but restoring *your* deleted or encrypted data is largely your responsibility (shared responsibility again). **Test a restore** on a schedule and record the result.

QUICK CHECK (answers at the back)

12. Why isn't the CVSS score alone enough to prioritise a fix?
13. What must you have before scanning a client's systems?

14. Why back up SaaS data such as Microsoft 365?

18 Running security monitoring as a service

> **Monitoring someone else's environment means turning alerts into clear, timely action, and showing the value in plain language.**

The operating loop

1. **Onboard:** agree scope in writing; install agents; connect log sources (endpoints, Microsoft 365 / Entra ID, firewall); record a baseline.
2. **Tune:** spend the first weeks removing noise and duplicates (your lab already taught you this).
3. **Triage:** agree response times per severity, e.g. "critical: acknowledged within 1 hour".
4. **Escalate and respond:** follow written playbooks; contact the agreed person; record every action.
5. **Report:** a short monthly report.
6. **Improve:** new detections from each incident and near-miss.

A monthly security report

Section	Contents
Summary	Three sentences: overall status, the biggest risk, the top recommendation
Alerts and incidents	Counts by severity; anything investigated and what was found
Vulnerabilities	Open critical/high items, and the trend versus last month
Hygiene	MFA coverage, patch status, backup test result, devices without protection
Recommendations	Up to three actions, ranked, in plain language

An incident response plan on one page

Who declares an incident; who to call (names and phone numbers, including outside help); first steps for common cases (phishing click, ransomware, account takeover, lost laptop); where backups are and how to restore; who handles legal, regulator and customer communication; and when to review the plan.

Professional responsibilities

- **A written agreement** covering scope, permissions, confidentiality (NDA), data handling and liability, before touching any system.
- **Handle client data carefully:** least access, encryption, and no client data in personal tools.
- **Know your limits:** have a plan for incidents beyond your capacity, such as an escalation partner.

QUICK CHECK (answers at the back)

15. What should be agreed in writing before monitoring a client?
16. List the five sections of a monthly security report.

19 Frameworks and compliance

> **Frameworks give a shared structure for security programmes; laws and standards set requirements organisations must meet. Knowing the main ones helps you speak the language of managers, auditors and clients.**

Name	Type	In one line
NIST Cybersecurity Framework 2.0	Framework	Six functions: **Govern, Identify, Protect, Detect, Respond, Recover**
CIS Critical Security Controls	Prioritised controls	18 controls; Implementation Group 1 is the essential baseline
ISO/IEC 27001	Certifiable standard	An information security management system (ISMS) with risk-based controls
SOC 2	Audit report	Common for service providers, especially selling to US clients
PCI DSS	Industry standard	Required for anyone storing, processing or transmitting payment card data
GDPR	Law (EU/UK)	Personal data protection, with breach notification duties
NDPA (Nigeria Data Protection Act 2023)	Law (Nigeria)	Personal data protection in Nigeria, overseen by the Nigeria Data Protection Commission (NDPC)
MITRE ATT&CK	Knowledge base	Attacker tactics and techniques, used to map detections (you already use it)

How they connect to your work

- Your Wazuh rules already show compliance mappings (PCI DSS, GDPR, NIST), because detection evidence supports audits.
- Cloud posture tools score you against benchmarks (CIS, the Microsoft cloud security benchmark, AWS best practices).
- Data protection laws usually require **appropriate security measures** and **timely breach notification**, which is why logging, IR plans and backups matter beyond technology.

REMEMBER

This guide explains frameworks for learning purposes. Legal requirements vary by country and change over time. For real compliance decisions, check the official text and get qualified legal advice.

QUICK CHECK (answers at the back)

17. Name the six functions of NIST CSF 2.0.
18. Which standard applies to organisations handling payment cards?
19. Which law governs personal data protection in Nigeria?

20 Glossary

Term	Meaning
Availability zone	A separate data centre within a region
BEC	Business email compromise: mailbox takeover or impersonation, often to redirect payments
Break-glass account	An emergency admin account, kept separate and closely monitored
CloudTrail	AWS log of API activity in an account
Conditional Access	Entra ID rules that grant or block sign-ins based on risk and context
CSPM	Cloud security posture management: continuous misconfiguration scanning
CVE / CVSS	A known vulnerability's ID / its severity score
DMARC	Email policy telling receivers how to handle unauthenticated mail
Entra ID	Microsoft's cloud identity service (formerly Azure AD)
GuardDuty	AWS threat detection service
IaC	Infrastructure as Code: defining infrastructure in files (e.g. Terraform)
KQL	Kusto Query Language, used in Sentinel and Log Analytics
Landing zone	A secure, pre-configured cloud foundation
Managed identity	An Azure identity for a resource, with no secrets to manage
NSG	Network security group: Azure's stateful network filter
PIM	Privileged Identity Management: just-in-time admin access
Private endpoint	A private IP inside your VNet for a PaaS service
RBAC	Role-based access control
SAS / pre-signed URL	A time-limited link granting access to storage
Sentinel	Microsoft's cloud SIEM and SOAR
Shared responsibility	The split of security duties between provider and customer
SOAR	Security orchestration, automation and response
VNet / VPC	A private network in Azure / AWS

Answers to the quick checks

1. On-demand self-service, broad network access, resource pooling, rapid elasticity, measured service.
2. Resource pooling.
3. You, the customer.
4. Azure: App Service or Azure SQL. AWS: Elastic Beanstalk or RDS.
5. Your data, your users and accounts, and access and configuration settings.
6. Private and public environments connected and used together.
7. Control over sensitive data, regulation, or predictable costs at large scale.
8. The provider.
9. Any two of: data, identities and accounts, access permissions, configuration.
10. An AWS account.
11. MFA on the top-level account, and a budget with alerts.
12. It has no stored secret that can leak; credentials are short-lived and managed for you.
13. Azure Bastion or AWS Session Manager.
14. A blob container inside a storage account.
15. Anyone who gets the link has access until it expires. Long-lived links act like leaked passwords.
16. Any two of: versioning, soft delete, immutable storage / Object Lock, backups.
17. The security group (network ACLs are stateless).
18. To reach a PaaS service through a private IP in your VNet, without using the public internet.
19. Shared security services are managed once, centrally, and every spoke's traffic can be controlled there.
20. Amazon SQS.
21. Any three of: no public access (private endpoints), identity-based authentication, encryption at rest and in transit, backups, auditing / threat detection, least-privilege database users.
22. A security principal, a role definition and a scope.
23. It's implicitly denied.
24. No long-lived secret to leak or rotate; Azure handles the credentials.
25. At rest (encryption, e.g. Key Vault / KMS keys); in transit (TLS / HTTPS-only); in use (confidential computing).
26. An encryption key you control in Key Vault or KMS. It lets you control rotation and revoke access, which some compliance rules require.
27. Azure Key Vault or AWS Secrets Manager.
28. Continuously scans cloud resources for misconfigurations and scores your security posture.
29. Any four of: public storage, SSH/RDP open to the internet, no MFA on admins, keys in code, over-permissive roles, public databases, logging off, public snapshots.
30. Any two of: it can be reviewed and scanned before deployment, it records changes in Git, it rebuilds environments identically, it destroys labs cleanly.
31. Azure: the Activity Log. AWS: CloudTrail.
32. CloudTrail events, VPC Flow Logs and DNS logs.
33. So an attacker who compromises a workload can't delete the evidence.
34. Preparation; detection and analysis; containment, eradication and recovery; post-incident activity.
35. Deactivate the key immediately (then investigate what it was used for).
36. To preserve evidence. Deleting the VM destroys its disk and the forensic trail.
37. MFA, patching, tested backups and email security.
38. Staff use standard accounts day to day, with admin rights only in separate accounts used when needed.
39. SPF lists allowed senders; DKIM signs messages; DMARC sets the policy for failures and collects reports.
40. To see all legitimate senders first, so real email isn't blocked when the policy is tightened.

41. Any two of: an unusual sign-in location, a new inbox rule, external forwarding, a sudden "change of bank details".
42. Severity doesn't show whether it's being exploited, whether the system is exposed, or how important the system is.
43. Written permission (part of the signed agreement).
44. Restoring your deleted or encrypted data is largely your responsibility, not the provider's.
45. Scope, permissions, confidentiality (NDA), data handling and liability.
46. Summary; alerts and incidents; vulnerabilities; hygiene; recommendations.
47. Govern, Identify, Protect, Detect, Respond, Recover.
48. PCI DSS.
49. The Nigeria Data Protection Act (NDPA) 2023, overseen by the NDPC.